

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: INDUSTRIAL AND SYSTEMS ENGINEERING (IM)

2. Subject Name: Supply Chain Management
Credits: 4

L-T-P: 3-1-0

3. Pre-requisites: OR I

4. Syllabus and reference books:

Syllabus: Introduction to Supply Chain Management, Supply Chain Structure and Strategic Choice, Supply Chain Processes and Decisions, Supply chain drivers and metrics

Distribution strategies, customer service, physical distribution planning, Material handling, Facility and warehousing decision, Supply Chain network design. Design of global supply chain network,

Planning and managing inventories in a supply chain, Coordinated inventory decisions, Product availability and Concept of safety stock, Fill rate measurement, Risk pooling, Quick response and Postponement strategy, Value of information sharing in supply chain management, Bullwhip effect in a supply chain.

Supply chain coordination, Concept of double marginalization, Supply chain contract design, Coordinated Pricing decision.

Sourcing decisions and outsourcing, the concept of total cost of ownership, Purchasing at Right Price, Supplier Relationships and Partnerships, Vendor Rating, Types of Contracts, Procurement Analytics, Supply Chain Risk Management, Understanding Risk in the SC and Risk Mitigation Strategies.

Sustainable Supply Chains, Concept of Triple Bottom-line, Closed loop supply chain IT in supply chain and its performance. Case Studies in supply chain management

Reference Books:

1) Supply Chain Management Strategy Planning and Operation by Sunil Chopra, Peter Meidl, Dharam Vir Kalra Publisher Pearson

2) Designing and managing the supply chain: Concepts, Strategies and Case studies by David Simchi Levi, Philip Kaminsky, Edith Simchi Levi published by Tata McGraw Hill Edition

3) Business Logistics/Supply Chain Management, Fifth Edition, Ronald H Ballaou, Samir K Srivastav, Pearson Education

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction to Supply Chain Management, Supply Chain Structure and Strategic Choice, Supply Chain Processes and Decisions, Supply chain drivers and metrics	6
2.	Distribution strategies, customer service, physical distribution planning, Facility and warehousing decision, Supply Chain network design. Design of global supply chain network,	10
3.	Planning and managing inventories in a supply chain, Coordinated inventory decisions, Product availability and Concept of safety stock, Fill rate measurement, Risk pooling, Quick response and Postponement strategy, Value of information sharing in supply chain management, Bullwhip effect in a supply chain.	10
4.	Supply chain coordination, Concept of double marginalization, Supply chain contract design, Coordinated Pricing decision.	6
5	Sourcing decisions and outsourcing, Supply Chain Risk Management, Understanding Risk in the SC and Risk Mitigation Strategies.	6
6	Sustainable Supply Chains, Concept of Triple Bottom-line, IT in supply chain and its performance.	4
Total number of hours		42